

Abdul Wali Khan University Mardan

Final Year Project Proposal Project

Title: " Machine Learning Based Music player app".

Student(s) Information:

Student 1

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Supervisor Information:

Name: -----

Designation: ----

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Project Introduction and problem statement:

This project aims to develop a *local music player app* that provides a lightweight, fast, and user-friendly interface for playing audio files directly from a user's phone storage.

The app will scan the device's local storage for music files and display them with relevant metadata such as title, etc. Users will be able to mark favorite songs, Shuffle, loop single, loop all and control playback speed, next, previous, also with features like repeat, all without an internet connection. The goal is to offer a smooth, distraction-free music experience with full control over the user's local music library.

Most modern music apps focus heavily on streaming and often require internet access, user accounts, or subscriptions. Users who prefer playing their *own locally stored music* often face:

- Lack of clean, modern apps for local playback
- Unwanted ads and online clutter
- Limited playlist and favorite management
- Poor UI and performance with large local libraries.
- Lack of self suggestion b/c of ML technology.

Project Domain :

Mobile Application Development.
Software Engineering.

Objectives:

1. Develop a user-friendly local music player app that scans and plays audio files from device local storage without requiring internet access.
2. Implement core playback features such as play, pause, next, previous , mark favorite songs, Shuffle, loop single, loop all and control playback speed, next, previous, also with features like repeat for seamless audio control.
3. Display song metadata (title, album,(other if necessary)) to enhance user experience and information visibility.
4. Add a favorites feature, so users can mark and quickly access their most-loved songs.
5. Ensure smooth performance across a variety of Android devices, even with large music libraries.
6. Maintain a lightweight and clean UI using Flutter, with focus on minimalism, usability, and responsiveness.
7. Self suggestion musics library based on multiple time played musics.

Scope of the Project:

1. UI where user can play, pause, repeat, delete, next previous, add/remove from favorite.

2, Favorites music list where user can access his favorite music with features like play, pause, repeat, next, previous, remove from favorite.

3, Self suggestion music library based on multiple times played music

Proposed Solution:

Technology : Flutter (Android).

Methodology: Agile approach with iterative design and testing.

Workflow:

App will access the music (.mp3 and audio more extensions) file and deliver to user with features like play, pause, repeat, delete, next, previous, add/remove from favorites.

Expected Outcomes:

Ads free.

Work offline.

Usability.

Minimum size on disk.

Attractive UI.

Self suggestion

Timeline (Milestones):

1. Requirement Analysis: 1 week.

2. UI : 2 weeks.

3. Implementation: 4 weeks.

4. Testing : 1 week

5. Documentation and submission : 1week

Resources Required:

Hardware & Software

Laptops (8GB+ RAM), test smartphones (Android)

Flutter SDK, Android Studio / VS Code

Figma / Canva for UI design

GitHub for version control, Trello/Jira for task tracking.

Risk Factors & Limitations:

Risk Factor:

1. For listening musics for long time in high volume in earphone may cause some ear related illnesses.
2. Limited time and expertise for advance features.
3. Only for android devices.

Limitations :

1. IOS support may be limited.
2. No safety for listening musics in high volume in earphone and.

Student Declaration:

We hereby declare that the proposed project is our original work and has not been copied from any source. All external sources, if used, will be properly cited in the final report.

Student Name:_____

Signature :_____.

Supervisor Signature :_____.